OPERATIVE REPORT COMPILATION

Patient:	Reyes, Yolanda
MRN:	RRTC-0074318
DOB:	06/22/1991 (34 yrs)
Sex:	F
Admitted:	10/18/2025
Attending of Record:	Priya Ramaswamy, MD — Trauma Surgery
Compilation printed:	11/17/2025

Contents: five operative reports (Trauma Surgery, Orthopedic Trauma Surgery, and Otolaryngology/General Surgery) for the 10/18/2025-11/05/2025 admission. Neurosurgical reports are filed separately by Rivergate Neurosurgical Consultants, Inc.

OPERATIVE REPORT — 1 of 5

Date of Procedure:	10/18/2025
Time in / out of room:	04:55 / 07:12
Location:	Trauma OR 1

PREOPERATIVE DIAGNOSIS

1. Hemoperitoneum with hemodynamic instability, mechanism blunt abdominal trauma (pedestrian struck by transit bus).
2. Grade IV splenic laceration with CT-documented active contrast extravasation.
3. Hemorrhagic shock with acute traumatic coagulopathy.
4. Open pelvic ring disruption (staged with orthopedic team, see companion report).

POSTOPERATIVE DIAGNOSIS

1. Shattered spleen (AAST Grade V at exploration) with hilar injury and ~2.5 L hemoperitoneum.
2. No hollow-viscus injury identified.
3. Extraperitoneal pelvic hematoma, contained, non-expanding at closure.
4. Intraperitoneal bladder-dome laceration, 2 cm, primarily repaired.
5. As stated above, continued shock physiology requiring damage-control closure.

PROCEDURES PERFORMED

1. Exploratory laparotomy via midline incision.
2. Total splenectomy for shattered spleen with hilar injury.
3. Ligation of short gastric vessels; mobilization of splenic flexure.
4. Primary repair of 2 cm intraperitoneal bladder-dome laceration in two layers.

5. Four-quadrant abdominal packing.
6. Damage-control temporary abdominal closure with negative-pressure ABTHERA dressing.

PERSONNEL

Surgeon: Priya Ramaswamy, MD — Trauma Surgery

First Assistant: Kwame Adekoya, MD — Trauma/Acute Care Surgery Fellow

Second Assistant: Delia Prasad, DO — PGY-4 General Surgery Resident

Urology (intraop consult, bladder repair): Isaac Nordstrom, MD

Anesthesia: General endotracheal (patient intubated in the field, ETT already in place); balanced anesthesia with fentanyl, midazolam, cisatracurium, low-dose isoflurane titrated to hemodynamics.

Anesthesiologist: Rania El-Sayegh, MD

INTRAOPERATIVE DATA

Estimated Blood Loss: ~3,000 mL intraoperatively (in addition to preoperative hemoperitoneum).

Fluids / Blood Products (intraop): 2 L crystalloid; per massive transfusion protocol — 12 units pRBC, 12 units FFP, 2 units apheresis platelets, 10 units cryoprecipitate; calcium gluconate ×3; tranexamic acid 1 g (loading dose in-flight, second gram intra-op).

Urine Output (intra-op): 180 mL, initially bloody, clearing.

Specimens to Pathology: Spleen, en bloc.

Cultures: Peritoneal fluid, aerobic and anaerobic.

Drains: Right upper quadrant Jackson-Pratt ×1; left upper quadrant Jackson-Pratt ×1; suprapubic drain adjacent to bladder repair.

Complications: None intraoperatively. Damage-control approach elected in view of ongoing coagulopathy, hypothermia (core 34.6 °C at start), and metabolic acidosis (pH 7.16, lactate 6.8 mmol/L at incision) — the classic “lethal triad.” Fascia intentionally left open.

INDICATIONS

Ms. Reyes is a 34-year-old previously healthy pedestrian who sustained polytrauma when struck by a transit bus at approximately 02:05 on 10/18/2025. Arrival trauma-bay FAST was positive in the left upper quadrant; CT confirmed a Grade IV splenic laceration with active contrast extravasation, moderate hemoperitoneum, an open pelvic ring disruption, and bilateral open tibia/fibula fractures. She was persistently hypotensive despite ongoing resuscitation under massive transfusion protocol, with a base deficit of –11 and lactate 6.4. Emergent operative source control was indicated. The plan discussed with the on-call orthopedic team was combined-case damage-control laparotomy followed immediately by external skeletal stabilization in the same anesthetic (see companion report).

FINDINGS

On entering the peritoneum, brisk hemoperitoneum was encountered, estimated at 2.5 L, immediately evacuated with pool suction. Systematic exploration revealed a shattered spleen with parenchymal disruption across all segments and a bleeding vascular pedicle at the hilum, consistent with AAST Grade V injury (worse than the CT-graded IV). No liver, pancreatic, gastric, small-bowel, colonic, or mesenteric injury was identified on repeated inspection. The retroperitoneum was carefully assessed: a zone II hematoma over the left kidney was non-expanding and not entered; a zone III pelvic hematoma was contained and non-expanding without evidence of arterial pulsation, consistent with the extraperitoneal bleeding expected from the pelvic ring injury and consistent with the orthopedic plan to compress the ring with an external fixator immediately following this case. A 2 cm laceration was identified on the dome of the bladder with clear extravasation of urine, correlating with the preoperative cystographic findings.

TECHNIQUE

The patient was already intubated and lines were in place from the trauma bay (right femoral cordis, bilateral 16-gauge peripheral IVs, right subclavian triple-lumen). She was positioned supine with arms out at 90° on padded arm boards.

The abdomen was prepped from the nipples to the mid-thighs with chlorhexidine and draped in standard sterile fashion, with the perineum prepped separately and draped off to preserve access for the orthopedic team.

A midline laparotomy incision was made from the xiphoid to the pubis with the No. 10 blade and carried through subcutaneous tissue and linea alba with electrocautery. Upon entering the peritoneum, brisk hemoperitoneum was encountered as described and evacuated. Four laparotomy pads were rapidly placed into each quadrant for temporary tamponade.

Attention was turned first to the left upper quadrant. The splenic flexure of the colon was mobilized and the spleen delivered into the wound. The splenophrenic, splenorenal, and splenocolic attachments were divided sharply and with electrocautery. The short gastric vessels were serially clamped, divided, and ligated with 2-0 silk ties. The splenic hilum was doubly ligated with 0 silk ties and transfixion-suture reinforced with 2-0 silk. The specimen was removed and passed off the field.

Careful inspection was then made of the pancreatic tail, which was uninjured. The stomach, duodenum, small bowel, and colon were run in their entirety without evidence of injury. The lesser sac was entered and the posterior stomach and body of the pancreas inspected — normal.

Attention was turned to the pelvis. The urinary bladder was distended with 300 mL of saline via the Foley to identify a 2 cm laceration at the dome, from which the instilled saline was seen to escape freely into the peritoneum. The laceration was debrided of devitalized edges and closed in two layers with 3-0 Vicryl (mucosa) and 2-0 Vicryl (seromuscular). Watertight closure was confirmed at 300 mL retrograde fill without leak. A suprapubic Jackson-Pratt drain was placed adjacent to the repair.

The zone III pelvic hematoma was inspected transperitoneally without entering it; it was contained and non-expansile. The retroperitoneum over the left kidney and the paracolic gutters were similarly inspected without entering the retroperitoneum. The mesentery, aorta, and IVC were palpated and appeared uninjured. The falciform ligament was taken down and the diaphragms and hepatic surfaces inspected — no injury.

In view of persistent coagulopathy (INR 1.9 at re-check), core temperature 34.8 °C intraoperatively, and continued acidosis, damage-control abdominal packing and temporary closure were elected. Four fresh laparotomy pads were placed in the splenic bed and pelvis. Two 19-Fr Jackson-Pratt drains were placed in the upper quadrants and brought out through separate stab incisions. An ABTHERA negative-pressure abdominal dressing was applied with the fenestrated visceral protective layer over the bowel, foam and adhesive drape overlying, and continuous suction at 125 mmHg. Fascia was intentionally left open, with planned second-look and definitive closure at 24–48 hours.

DISPOSITION

Patient remained intubated and was transferred, still under general anesthesia, to the adjacent OR position for orthopedic external-fixator placement (see companion report). Post-orthopedic case, she was transferred directly to the Surgical/Neurosurgical ICU on norepinephrine, propofol/fentanyl sedation, continued blood-product resuscitation, ICP monitor placed by neurosurgery at the ICU bedside. Family (sister Marisela Reyes) present in surgical waiting; updated in person by the trauma team.

Electronically signed by:

Priya Ramaswamy, MD

Attending, Trauma Surgery

Rivergate Regional Trauma Center

10/18/2025 09:14

OPERATIVE REPORT — 2 of 5

Date of Procedure: 10/18/2025
Time in / out of room: 07:20 / 09:05 (combined-case, same anesthetic as Op Report 1)
Location: Trauma OR 1

PREOPERATIVE DIAGNOSIS

1. Open (Gustilo IIIA) vertically unstable left-sided pelvic ring disruption: displaced fractures of the left superior and inferior pubic rami with left sacroiliac joint diastasis; open wound over left iliac crest.
2. Right open comminuted mid-shaft tibia and fibula fractures (Gustilo IIIA).
3. Left open comminuted distal tibia and fibula fractures (Gustilo IIIB), with soft-tissue loss over the anteromedial ankle.
4. Polytrauma with damage-control priorities; hemodynamic instability.

POSTOPERATIVE DIAGNOSIS

Same as preoperative.

PROCEDURES PERFORMED

1. Anterior pelvic external fixator, two supra-acetabular pins per side, connected via anterior bar construct.
2. Right lower extremity spanning external fixator, knee-to-hindfoot, with pins in distal femur, proximal tibia, and calcaneus, bridging the tibial shaft fracture.
3. Left lower extremity spanning external fixator, knee-to-hindfoot, bridging the distal tibial fracture, with pins in distal femur, proximal tibia, and calcaneus.
4. Irrigation and debridement of all three open fracture wounds (left pelvic/iliac crest wound, right mid-tibial wound, left anteromedial ankle wound), with pulsed-lavage 9 L normal saline per site.
5. Application of antibiotic bead pouches (vancomycin/tobramycin PMMA beads) to the left tibia/ankle wound bed.
6. Placement of negative-pressure wound therapy dressings (Wound VAC) over all three open fracture wounds.

PERSONNEL

Surgeon: Kavita Sundararajan, MD — Orthopedic Trauma Surgery

First Assistant: Timothy Björkman, DO — PGY-5 Orthopedic Surgery Chief Resident

Second Assistant: Anselm Vukovic, MD — PGY-3 Orthopedic Surgery Resident

Anesthesia: Continuation of general endotracheal anesthesia from combined trauma-surgery case (see Op Report 1).

Anesthesiologist: Rania El-Sayegh, MD

INTRAOPERATIVE DATA

Estimated Blood Loss: ~450 mL (orthopedic portion only; total combined-case EBL documented in trauma-surgery report).

Fluids / Blood Products: Continuation from prior case; additional 2 units pRBC, 2 units FFP given during orthopedic portion.

Specimens to Pathology: None (bone/tissue debrided but not submitted).

Cultures: Deep-wound aerobic and anaerobic cultures from each of the three open fracture sites.

Complications: None intraoperatively.

Antibiotic Prophylaxis: IV cefazolin 2 g on-call to OR; IV gentamicin 5 mg/kg per open-fracture protocol; local vancomycin/tobramycin PMMA beads at Gustilo IIIB site.

Tetanus: Tdap administered in trauma bay.

Imaging: Intraoperative C-arm fluoroscopy throughout.

INDICATIONS

Vertically unstable open pelvic ring injury and bilateral open comminuted tibia/fibula fractures in the setting of polytrauma including severe TBI and post-laparotomy damage-control state. External skeletal stabilization was indicated to (1) achieve mechanical stability and hemostatic compression of the pelvic ring, thereby limiting ongoing venous and cancellous blood loss into the retroperitoneum, and (2) span and provisionally stabilize the bilateral tibial fractures while allowing serial washouts and soft-tissue management before staged definitive fixation. Consent obtained via two-physician emergency waiver in the setting of surrogate unavailability at the moment of decision-making; family (sister) notified as soon as reachable.

FINDINGS

- **Pelvis:** Vertically unstable left-sided pelvic ring injury with 2.8 cm diastasis at the left sacroiliac joint on preoperative CT and comminution of the left superior and inferior pubic rami. The open wound over the left iliac crest measured approximately 4 cm and tracked deep to the crest with visible cancellous bone. Contamination with roadway grit was noted.
- **Right tibia/fibula:** Segmental mid-shaft tibial fracture with a butterfly fragment; associated mid-shaft fibula fracture. Open wound anteriorly, ~8 cm, Gustilo IIIA (adequate soft-tissue coverage of exposed bone after debridement).
- **Left tibia/fibula:** Comminuted distal tibial pilon-pattern fracture with an associated distal fibula fracture. Anteromedial ankle wound ~10 cm with significant soft-tissue loss and denuded distal tibia — Gustilo IIIB with anticipated need for staged plastic surgery coverage.
- Neurovascular status distally: dorsalis pedis pulses palpable bilaterally by handheld Doppler after stabilization; compartments soft bilaterally.

TECHNIQUE

With the patient supine on the flat-top radiolucent OR table (positioning maintained from the trauma-surgery case, with abdominal ABTHERA in place), the pelvis and both lower extremities were re-prepped with chlorhexidine and re-draped for the orthopedic case. The C-arm was brought in from the patient's right for pelvic imaging.

Anterior pelvic external fixator. Under fluoroscopic guidance in inlet and outlet views, two supra-acetabular pins (5 mm hydroxyapatite-coated Schanz pins) were placed per side via 1.5 cm stab incisions two fingerbreadths above and lateral to the anterior inferior iliac spine, directed toward the posterior superior iliac spine. Pin position and length were confirmed on fluoroscopy. A carbon-fiber bar was mounted to the pins with clamps, the ring reduced by manual compression from the assistants, and the construct locked with the diastasis reduced to <5 mm on the outlet view.

Left iliac-crest wound washout. The wound was extended for exposure, foreign body (roadway debris, grit) removed, devitalized skin, subcutaneous fat, and periosteum sharply debrided. Pulsed lavage with 3 L normal saline. Deep culture obtained. Wound left open, packed with saline-moistened gauze pending Wound VAC placement.

Right lower-extremity spanning fixator. The right open tibial wound was extended proximally and distally for exposure. Foreign material irrigated out, devitalized soft tissue and small avascular bone fragments sharply debrided. Pulsed lavage with 3 L normal saline. Deep culture obtained. The wound was left open. Distal femoral, proximal tibial, and calcaneal Schanz pins were placed via stab incisions under fluoroscopic guidance and connected across the knee and ankle with carbon-fiber bars. Length, alignment, and rotation were restored and confirmed fluoroscopically.

Left lower-extremity spanning fixator. The left anteromedial ankle wound was extended for exposure. Extensively contaminated. Foreign material removed. Devitalized skin, subcutaneous tissue, and denuded periosteum sharply debrided. Pulsed lavage with 3 L normal saline. Deep culture obtained. Vancomycin/tobramycin PMMA antibiotic beads (six beads on braided suture) were placed in the tibial defect. Distal femoral, proximal tibial, and calcaneal Schanz pins were placed and connected across knee and ankle with carbon-fiber bars, restoring length, alignment, and rotation under fluoroscopy.

Wound VAC dressings were applied to all three open wounds at 125 mmHg continuous suction. Pin sites were dressed with iodine-impregnated gauze.

Final fluoroscopic images of the pelvis (AP, inlet, outlet) and bilateral tibiae (AP, lateral) documented acceptable provisional reduction and hardware position.

DISPOSITION

Patient remained intubated and hemodynamically supported. Transferred directly from the OR to the Surgical/Neurosurgical ICU, still under general anesthesia, on norepinephrine and continued blood-product resuscitation. Neurosurgery placed an intraparenchymal ICP monitor at the ICU bedside upon arrival. Plan for return to OR within 48–72 hours for second-look laparotomy and abdominal closure, and for staged serial orthopedic washouts (see Op Report 3).

Electronically signed by:

Kavita Sundararajan, MD

Attending, Orthopedic Trauma Surgery

Rivergate Regional Trauma Center

10/18/2025 10:22

OPERATIVE REPORT — 3 of 5

Date of Procedure:	10/22/2025
Time in / out of room:	08:10 / 10:35
Location:	Trauma OR 3

PREOPERATIVE DIAGNOSIS

1. Status-post external fixation, open Gustilo IIIA left pelvic ring / iliac crest wound (POD #4).
2. Status-post external fixation, right open Gustilo IIIA mid-shaft tibia/fibula fracture (POD #4).
3. Status-post external fixation, left open Gustilo IIIB distal tibia/fibula fracture with soft-tissue loss (POD #4); antibiotic-bead pouch in situ.
4. Serial washout indicated per open-fracture protocol.

POSTOPERATIVE DIAGNOSIS

Same. Wounds showing appropriate granulation without gross evidence of deep infection; left ankle wound with persistent moderate soft-tissue defect anticipated to require plastic-surgery flap coverage at time of definitive fixation.

PROCEDURES PERFORMED

1. Serial washout and irrigation, left iliac-crest open pelvic wound.
2. Serial washout and irrigation, right open tibia/fibula wound.
3. Serial washout and irrigation, left open tibia/fibula/ankle wound; exchange of vancomycin/tobramycin PMMA antibiotic beads.
4. Repeat sharp debridement of any remaining devitalized tissue at each site.
5. Wound VAC exchange, all three sites.
6. Repeat deep-wound cultures at each site.

PERSONNEL

Surgeon: Kavita Sundararajan, MD — Orthopedic Trauma Surgery

First Assistant: Timothy Björkman, DO — PGY-5 Orthopedic Surgery Chief Resident

Plastic Surgery (intraop consult, left ankle only): Yuki Hasegawa, MD

Anesthesia: General endotracheal (patient still intubated in ICU).

Anesthesiologist: Marcus Odell, MD

INTRAOPERATIVE DATA

Estimated Blood Loss: ~150 mL total across three sites.

Fluids: 1,200 mL crystalloid; no blood products required.

Specimens to Pathology: None.

Cultures: Deep-wound aerobic and anaerobic cultures from each open-fracture site. Prior 10/18 cultures reported no growth to date at 96 hours.

Complications: None.

Antibiotics: IV cefazolin re-dosed on-call to OR; local vancomycin/tobramycin PMMA beads re-exchanged at left ankle wound.

INDICATIONS

Standard-of-care serial washout every 48–72 hours for Gustilo grade III open fractures until definitive fixation and/or soft-tissue coverage. Between 10/18 and 10/22, patient tolerated resuscitation with successful damage-control laparotomy closure on 10/20 (documented separately), gradually weaning vasopressor support, and stable ICP with EVD/monitor in place. She remained intubated with a Richmond Agitation-Sedation Scale (RASS) of –4 to –5 during this period.

FINDINGS

- **Left iliac-crest wound:** Clean granulating base, no purulence, no foul odor. Cancellous bone edges viable.
- **Right tibia wound:** Beefy granulation tissue at the base; small non-viable skin edges circumferentially. Bone exposed centrally at fracture site, cortex viable and bleeding on pinprick.
- **Left ankle wound:** Larger residual soft-tissue defect anteromedially as previously documented; base clean, granulating; distal tibia cortex partially covered by early granulation; antibiotic beads intact. Plastic surgery evaluated intraoperatively and confirmed candidacy for local rotational or free-tissue flap coverage at definitive-fixation stage.

TECHNIQUE

After general anesthesia (continued from ICU), the patient was positioned supine and all pin sites and dressings were carefully unwrapped. The three Wound VACs were disconnected and removed sequentially. Each site was inspected as detailed above.

At each site, sharp debridement of any remaining devitalized skin edges was performed. Pulsed lavage with 6 L normal saline per site was carried out. Deep tissue and bone cultures were obtained.

At the left ankle wound, the previously placed antibiotic-bead string was removed and passed off the field. A fresh six-bead vancomycin/tobramycin PMMA string was fashioned on braided nonabsorbable suture and placed into the tibial defect.

Fresh Wound VAC dressings were applied at 125 mmHg continuous suction across all three sites. Pin sites were cleaned with chlorhexidine and re-dressed with iodine-impregnated gauze. External-fixator constructs were inspected and confirmed stable, with all clamps re-tightened.

DISPOSITION

Extubation not attempted; patient returned intubated to the Surgical/Neurosurgical ICU on continued sedation. Plan: definitive ORIF pelvis and both tibiae on 10/25/2025, with plastic surgery participating in left ankle soft-tissue coverage (see Op Report 4).

Electronically signed by:

Kavita Sundararajan, MD

Attending, Orthopedic Trauma Surgery

Rivergate Regional Trauma Center

10/22/2025 11:18

OPERATIVE REPORT — 4 of 5

Date of Procedure:	10/25/2025
Time in / out of room:	07:45 / 15:20
Location:	Trauma OR 1

PREOPERATIVE DIAGNOSIS

1. Open (Gustilo IIIA) unstable left-sided pelvic ring disruption: left superior/inferior pubic rami fractures with left sacroiliac joint diastasis; wound now clean and granulating (POD #7).

2. Right open Gustilo IIIA mid-shaft tibia/fibula fracture, spanning external fixator in place (POD #7).

3. Left open Gustilo IIIB distal tibial pilon-pattern fracture with associated distal fibula fracture and residual anteromedial soft-tissue defect (POD #7).

POSTOPERATIVE DIAGNOSIS

Same, now with definitive internal fixation and soft-tissue coverage as described.

PROCEDURES PERFORMED

1. Open reduction and internal fixation, left pelvic ring: percutaneous iliosacral screw fixation of left sacroiliac joint diastasis; anterior symphyseal plating with 6-hole reconstruction plate spanning the pubic rami; removal of anterior pelvic external fixator.

2. Open reduction and internal fixation, right mid-shaft tibia: reamed intramedullary interlocking nail; ORIF fibula not required (acceptable alignment).

3. Open reduction and internal fixation, left distal tibia (pilon): medial locking plate with anatomically contoured distal tibia plate; independent lag screw fixation of articular fragments; open reduction and internal fixation of left distal fibula with one-third tubular plate.

4. Removal of bilateral lower-extremity spanning external fixators.

5. Removal of antibiotic-bead pouch, left ankle.

6. Left ankle soft-tissue coverage: pedicled medial gastrocnemius rotational flap with split-thickness skin graft (harvested from ipsilateral anterolateral thigh) over the flap and residual defect — performed by Plastic Surgery.

7. Application of clean postoperative dressings; short-leg posterior splint applied on left; long-leg posterior splint applied on right.

PERSONNEL

• **Orthopedic Trauma:** Kavita Sundararajan, MD (attending); Timothy Björkman, DO (PGY-5 chief resident, first assistant).

• **Plastic Surgery (left ankle flap and STSG):** Yuki Hasegawa, MD (attending); Rosalind Efobi, MD (PGY-4 resident).

Anesthesia: General endotracheal (patient remained intubated preoperatively).

Anesthesiologist: Rania El-Sayegh, MD

INTRAOPERATIVE DATA

Estimated Blood Loss: ~1,150 mL (combined orthopedic + plastic-surgery portions).

Fluids / Blood Products: 3,500 mL crystalloid; 2 units pRBC.

Specimens to Pathology: Portions of resected devitalized soft tissue; explanted antibiotic beads (not to pathology, gross log only).

Cultures: Deep-wound aerobic and anaerobic cultures at each site prior to hardware placement.

Complications: None intraoperative. Case duration extended by combined orthopedic-plastic sequence.

Antibiotics: IV cefazolin 2 g on-call and re-dosed q4h intra-op; single dose IV vancomycin 1 g given plastic-surgery involvement.

Imaging: Intraoperative C-arm fluoroscopy for pelvis and both tibiae; final AP and lateral fluoroscopic images retained.

INDICATIONS

Patient POD #7 from initial injury with polytrauma stabilized. Neurologic status improved from initial GCS 5T to GCS 7T (still sedated); ICP consistently <20 mmHg for >72 hours off osmotherapy; hemodynamically stable off vasopressors; abdomen closed on 10/20; nutritional support via NG feeding tolerated. Definitive skeletal fixation now indicated and safely feasible ("scanning-window" criteria met). Combined case coordinated with plastic surgery for left ankle soft-tissue coverage in the same anesthetic.

FINDINGS

- **Pelvis:** Left SI joint diastasis reduced to <2 mm under manual compression via anterior fixator; sacroiliac screw trajectory clean on inlet/outlet/lateral views. Symphysis diastasis 1.5 cm, plated in anatomic reduction. Iliac-crest wound base clean, granulating, closed primarily over a drain.
- **Right tibia:** Segmental mid-shaft fracture with butterfly fragment as previously documented; reduction achieved on traction with restoration of length, alignment, and rotation on fluoroscopy. Wound base clean.
- **Left tibia (pilon):** Comminution of the distal metaphysis and articular surface with a large medial fragment, a posterior malleolar equivalent, and a central impaction fragment. Articular reduction achieved and maintained with independent 3.5 mm lag screws; medial locking plate spans metaphyseal comminution. Fibula reduced and stabilized with one-third tubular plate.
- **Left ankle soft-tissue defect:** Successfully covered by pedicled medial gastrocnemius rotational flap; well-perfused at closure; STSG (0.014" thickness) applied over flap and adjacent defect and secured with staples and bolster dressing.

TECHNIQUE (SUMMARY)

Pelvis. Patient supine on radiolucent table with fluoroscopy from the right. External fixator loosened but maintained for provisional reduction. A 1 cm stab incision made lateral to the left PSIS. Under multiplanar fluoroscopy (inlet, outlet, and true lateral sacral views), a guide wire was advanced across the sacroiliac joint into the S1 body, confirmed intraosseous on all three views. A 7.3 mm cannulated partially threaded screw with washer, 90 mm in length, was placed over the guide wire and compressed the SI joint. A second S1 screw of 85 mm was placed as a fully threaded position screw for additional stability. Anterior symphyseal plating was performed via a Pfannenstiel incision. The rectus insertion was elevated in a sub-periosteal fashion. The symphysis was reduced with a Weber clamp. A 6-hole 3.5 mm curved reconstruction plate was contoured and secured with three 3.5 mm cortical screws per side. Reduction confirmed on fluoroscopy. Anterior pelvic external fixator was removed. Wound closed in layers over a Blake drain. The iliac-crest wound was closed primarily with 2-0 Vicryl and staples after further granulation-tissue debridement.

Right tibia. External fixator removed. Standard suprapatellar semi-extended approach for tibial nail. Entry point established at anterior tibial cortex just posterior to the patellar tendon insertion. Reamer sequence to 11.5 mm. A 10 mm × 360 mm tibial nail was placed with two proximal and two distal interlocking screws under fluoroscopy. Length, alignment, and rotation confirmed. Anterior open wound irrigated and closed primarily over a Penrose drain after further granulation-edge debridement.

Left tibia (pilon) and fibula. External fixator removed. Anteromedial approach to distal tibia, extended from prior wound. Articular fragments identified and reduced with pointed reduction clamps; independent 3.5 mm lag screws placed across articular fragments. Medial locking plate (distal tibia anatomic) applied and secured with locking screws proximally and distally per templating. Lateral approach to distal fibula; standard ORIF with one-third tubular plate and 3.5 mm cortical screws.

Left ankle soft-tissue coverage (Plastic Surgery). Medial gastrocnemius muscle harvested via posteromedial calf incision, elevated on its medial sural vessels, and rotated into the anteromedial ankle defect with tension-free coverage. Muscle secured with 3-0 Vicryl. STSG harvested from ipsilateral anterolateral thigh with dermatome at 0.014"; meshed 1.5:1 and applied over the flap and residual periphery of the defect; secured with staples and a tie-over bolster. Donor site dressed with Xeroform and dry gauze.

Final fluoroscopy documented acceptable pelvic reduction and hardware position; anatomic tibial and fibular alignment bilaterally.

DISPOSITION

Patient remained intubated and returned to the Surgical/Neurosurgical ICU for postoperative recovery. Postoperative weight-bearing status: non-weight-bearing bilateral lower extremities; touch-down weight-bearing pelvis when eventually mobilized. Bolster dressing over left ankle STSG to be taken down at POD #5.

Electronically signed by:

Kavita Sundararajan, MD — Orthopedic Trauma Surgery — 10/25/2025 16:47

Yuki Hasegawa, MD — Plastic and Reconstructive Surgery — 10/25/2025 16:52

OPERATIVE REPORT — 5 of 5

Date of Procedure:	11/05/2025
Time in / out of room:	10:15 / 11:48
Location:	Trauma OR 4

PREOPERATIVE DIAGNOSIS

1. Prolonged respiratory failure requiring continued mechanical ventilation, HD #19 of translaryngeal intubation.

2. Severe traumatic brain injury with diffuse axonal injury; persistent depressed level of consciousness (current GCS approximately 8T off sedation).

3. Anticipated prolonged inability to protect airway and to safely take oral nutrition.

4. Need for durable enteral access for nutrition and medication administration.

POSTOPERATIVE DIAGNOSIS

Same.

PROCEDURES PERFORMED

1. Open surgical tracheostomy with placement of size 8 Shiley cuffed nonfenestrated tracheostomy tube.

2. Percutaneous endoscopic gastrostomy (PEG) tube placement, size 20 Fr, via pull technique.

PERSONNEL

• **Tracheostomy (Otolaryngology — Head & Neck Surgery):** Emilia Vaccaro, MD (attending); Julien Marchetti, MD (PGY-4 resident, first assistant).

• **PEG (General Surgery):** Alonzo Kirkham, MD (attending); Delia Prasad, DO (PGY-4 resident, first assistant).

Anesthesia: General endotracheal (patient remained intubated); converted to tracheostomy tube ventilation intraoperatively during the tracheostomy portion.

Anesthesiologist: Marcus Odell, MD

INTRAOPERATIVE DATA

Estimated Blood Loss: <25 mL (tracheostomy); <10 mL (PEG). Combined <50 mL.

Fluids: 800 mL crystalloid.

Specimens to Pathology: None.

Cultures: None obtained.

Complications: None intraoperatively.

Antibiotics: IV cefazolin 2 g on-call; single-dose cefazolin appropriate for PEG placement.

INDICATIONS

Ms. Reyes has now completed 19 days of translaryngeal intubation following her polytrauma admission on 10/18/2025. She has undergone successful damage-control laparotomy with subsequent abdominal closure, external and then definitive internal fixation of an open pelvic ring injury and bilateral open tibia/fibula fractures, and left-ankle soft-tissue coverage with a rotational flap and STSG. Neurologic recovery is slow: she now opens eyes to voice intermittently and localizes with the right upper extremity, but does not follow commands consistently and does not protect her airway; she has failed two spontaneous breathing trials over the past week with poor cough and copious secretions. Continued long-term ventilatory and nutritional support is anticipated. Tracheostomy and PEG are indicated for airway hygiene, ventilator weaning, prevention of laryngeal injury from prolonged translaryngeal intubation, and reliable enteral access. Risks, benefits, and alternatives were discussed with the surrogate decision-maker (sister Marisela Reyes, temporary conservator per court order dated 11/10/2025 — pending at time of scheduling and confirmed by 11/12/2025; consent obtained via two-physician documentation of surgical necessity in the interim, with subsequent conservator ratification on file); documented in the medical record.

FINDINGS

• **Tracheostomy:** Normal midline anatomy; palpable cricoid and tracheal rings; no aberrant vasculature identified. Trachea entered between the second and third tracheal rings without difficulty. Existing endotracheal tube withdrawn to the level of the vocal cords under direct visualization; tracheostomy tube advanced into the trachea and connected to the ventilator circuit with prompt return of end-tidal CO■ and bilateral breath sounds.

• **PEG:** Diagnostic esophagogastroduodenoscopy performed to the second portion of the duodenum; no mucosal abnormalities of the esophagus, stomach, or duodenum. Adequate transillumination of the anterior abdominal wall in the left upper quadrant; one-to-one indentation confirmed. Prior midline laparotomy incision well healed with mature scar; no interference with the selected PEG site.

TECHNIQUE

Tracheostomy. With the patient supine, a shoulder roll was placed to extend the neck. The anterior neck was prepped with chlorhexidine and draped in standard sterile fashion. Anatomic landmarks — thyroid notch, cricoid cartilage, sternal notch — were identified and marked. A 3 cm transverse incision was made two fingerbreadths above the sternal notch and carried through subcutaneous tissue and platysma with electrocautery. Strap muscles were separated in the midline and retracted laterally. The thyroid isthmus was identified, divided with ligation using 2-0 silk, and reflected to expose the pretracheal fascia. The trachea was identified and confirmed by needle aspiration of air. Anesthesia was notified, FiO■ increased to 1.0, and the endotracheal tube was withdrawn under bronchoscopic guidance to the level of the vocal cords. A vertical incision was made between the second and third tracheal rings and an inferiorly based Björk flap fashioned and secured to the inferior skin edge with 3-0 Vicryl. A size 8 Shiley cuffed nonfenestrated tracheostomy tube was inserted, the obturator removed, and the inner cannula placed. The cuff was inflated and the circuit connected with immediate return of end-tidal CO■ and equal bilateral breath sounds. The tube was secured with sutures at four corners and a tracheostomy collar.

PEG. Attention was turned to the abdomen. The abdomen was prepped and draped in standard sterile fashion. The gastroenterologist (see attending above, credentialed in endoscopy) advanced the gastroscope through the oropharynx into the esophagus, stomach, and duodenum, then withdrew to the body of the stomach with insufflation. The room lights were dimmed and adequate transillumination visualized in the left upper quadrant, well clear of the healed midline laparotomy scar and any drain sites. A finger-indentation test confirmed one-to-one correspondence.

The selected site was infiltrated with 1% lidocaine with epinephrine and a small stab incision made. Under endoscopic visualization, a needle-cannula was passed through the abdominal wall into the stomach, the inner needle withdrawn, and a looped wire advanced through the cannula and grasped endoscopically. The wire was withdrawn through the mouth with the endoscope. A 20 Fr pull-type PEG tube was attached to the wire at the mouth and pulled retrograde through the esophagus, stomach, and out through the abdominal wall, seating the internal bumper against the gastric mucosa under repeat endoscopic visualization at 3 cm depth. The external bolster was secured at 4 cm without excessive tension. Gastroscope removed. PEG capped and dressed.

DISPOSITION

Patient returned to the Surgical/Neurosurgical ICU. Ventilator settings continued on assist-control mode via tracheostomy. PEG to gravity drainage ×24 hours, then tube-feed initiation per critical-care nutrition protocol. Plan for weaning trials via tracheostomy collar; discharge planning to inpatient long-term acute care / rehabilitation once medically ready.

Electronically signed by:

Emilia Vaccaro, MD — Otolaryngology–Head & Neck Surgery — 11/05/2025 12:35

Alonzo Kirkham, MD — General Surgery — 11/05/2025 12:41

End of operative report compilation for Reyes, Yolanda (MRN RRTC-0074318), covering 10/18/2025 through 11/05/2025. Additional operative or bedside procedures performed after 11/05/2025 documented separately in the medical record.